Noether's Theorem

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1 Introduction

Noether's theorem states that every continuous symmetry of the action associates with a conserved current. In this note, we provide a precise statement and a proof of Noether's theorem for classical fields under Lagrangian formulation.

2 Noether's Theorem

2.1 Group Action on Fields

For simplicity, we consider only vector fields, the idea is easily extended to other fields. Let $\mathbb{R}^4$ be the Minkowski space and V be a vector space. A field ϕ is a smooth map from $\mathbb{R}^4$ to V. Let G be a group, $A : G \to \mathrm{GL}(V)$, $B : G \to \mathrm{Diff}(\mathbb{R}^4)$ be two group homomorphisms, where $\mathrm{GL}(V)$ is the group of invertible linear maps on V, $\mathrm{Diff}(\mathbb{R}^4)$ is the group of diffeomorphisms of $\mathbb{R}^4$. Let $g \in G$, the field $g\phi$ is then defined as

$$(g\phi)(p) \equiv A(g)\phi(B(g^{-1})p). \tag{1}$$

Under this definition, we have $g(h\phi) = (gh)\phi$. When there is no ambiguity, we will write

$$(g\phi)(p) \equiv g(\phi(g^{-1}p)). \tag{2}$$

2.2 The Lagrangian and The Action

The Lagrangian is a map mapping a field ϕ to another field $\mathcal{L}[\phi]$. Due to the locality of physical laws, we consider the special case where there is a function L such that the value of $\mathcal{L}[\phi]$ at a point p satisfies

$$\mathcal{L}[\phi](p) = L(\phi(p), (\partial_\mu\phi)(p)). \tag{3}$$

The action $\mathcal{S}$ is then a map mapping a compact submanifold $M \subset \mathbb{R}^4$ together with a field ϕ to $\mathbb{R}$, defined as

$$\mathcal{S}(M, \phi) \equiv \int_M L(\phi(p), (\partial_\mu\phi)(p))dV, \tag{4}$$

where dV is the volume form on $\mathbb{R}^4$ under the Minkowski metric.

2.3 Symmetry of The Action

Definition 2.1. *Let G be a group, G is said to be a symmetry group of the action $\mathcal{S}$ if there exists functions $F^\mu : V \to \mathbb{R}$ such that for all $g \in G$ and all field ϕ, we have*

$$\mathcal{S}(gM, g\phi) + \int_{gM} \partial_\mu[F^\mu((g\phi)(p), (\partial_\nu(g\phi))(p)))]dV = \mathcal{S}(M, \phi) + \int_M \partial_\mu[F^\mu(\phi(p), (\partial_\nu\phi)(p)))]dV. \tag{5}$$

Remark 1: Naively we may want to define a symmetric operation g of the action to satisfy $\mathcal{S}(gM, g\phi) = \mathcal{S}(M, \phi)$. Indeed, Eq. (5) is essentially doing the same thing. Since adding a divergence term in the action would not change the equation of motion, we can define a new action $\mathcal{S}_{new}$ to be

$$\mathcal{S}_{new}(M, \phi) = \mathcal{S}(M, \phi) + \int_M \partial_\mu[F^\mu(\phi(p), (\partial_\nu\phi)(p))]dV. \tag{6}$$

$\mathcal{S}$ and $\mathcal{S}_{new}$ describe the same physical system. For $\mathcal{S}_{new}$, we have

$$\mathcal{S}_{new}(gM, g\phi) = \mathcal{S}_{new}(M, \phi). \tag{7}$$

In other words, the symmetry group defined in Definition 2.1 is not a symmetry of the action $\mathcal{S}$ but a symmetry of the physical system described by $\mathcal{S}$.

Remark 2: The function F^μ may depend only on both field value and its derivatives. All we need is that there exists an $\mathcal{S}$ whose Lagrangian takes the form of Eq. (4) to guarantee that the equation of motion depends at most on second order derivatives.

2.4 Noether's Theorem: The Statement and The Derivation

Now we are ready to state the **Noether's theorem**.

Theorem 2.1. *Let G be a Lie group with generators $Q_1, ..., Q_n$, and $\mathcal{S}$ be an action of some physical system. If G is a symmetry group of the action $\mathcal{S}$, then for all fields ϕ satisfying the Euler-Lagrangian equation, we have conserved currents J_i^μ*

$$\begin{aligned} &J_i^\mu \equiv L\xi_i^\mu + \frac{\partial L}{\partial(\partial_\mu\phi)} Q_i\phi + \frac{\partial F^\mu}{\partial\phi} Q_i\phi + \frac{\partial F^\mu}{\partial(\partial_\nu\phi)}\partial_\nu\left(Q_i\phi\right) + \xi_i^\mu \partial_\nu F^\nu, \\ &\text{where } \xi_i^\mu(p) \equiv \lim_{\epsilon\to 0}\frac{x^\mu(e^{Q_i\epsilon}p) - x^u(p)}{\epsilon}, (Q_i\phi)(p) \equiv \lim_{\epsilon\to 0}\frac{(e^{Q_i\epsilon}\phi)(p) - \phi(p)}{\epsilon} \end{aligned} \tag{8}$$

Here x^μ are the coordinate functions and p is a point in $\mathbb{R}^4$.

Remark 3: J_i^μ being a conserved current is equivalent to saying that $\partial_\mu J_i^\mu = 0$. Note this equation has the same form with the charge conservation equation, this is where the name "conserved current" comes from.

Proof. For simplicity, we assume that G is a 1 dimensional Lie group with generator Q, i.e., any element $g \in G$ takes the form $g = e^{Qt}$ for some $t \in \mathbb{R}$. We define

$$(R\phi)(p) \equiv \lim_{\epsilon\to 0}\frac{A(e^{Q\epsilon})(\phi(p)) - \phi(p)}{\epsilon} \tag{9}$$

First, up to terms of order $\mathcal{O}(\epsilon^2)$, using

$$\lim_{\epsilon\to 0}\frac{(\partial_\mu(e^{Q\epsilon}\phi))(e^{Q\epsilon}p)) - (\partial_\mu\phi)(p))}{\epsilon} = R\partial_\mu\phi(p) - \partial_\mu\xi^\nu(p)\partial_\nu\phi(p), \tag{10}$$

we have

$$\begin{aligned} &\int_{e^{Q\epsilon}M} L((e^{Q\epsilon}\phi)(p), (\partial_\mu(e^{Q\epsilon}\phi))(p))dV - \int_M L(\phi(p), (\partial_\mu\phi)(p))dV \\ =&\int_M (e^{Q\epsilon})^*[L((e^{Q\epsilon}\phi)(p), (\partial_\mu(e^{Q\epsilon}\phi))(p))dV] - \int_M L(\phi(p), (\partial_\mu\phi)(p))dV \\ =&\int_M L((e^{Q\epsilon}\phi)(e^{Q\epsilon}p), (\partial_\mu(e^{Q\epsilon}\phi))(e^{Q\epsilon}p))\det(J(e^{Q\epsilon})(p))dV - \int_M L(\phi(p), (\partial_\mu\phi)(p))dV \\ =&\epsilon\left(\int_M L\partial_\mu\xi^\mu dV + \int_M\left(\frac{\partial L}{\partial\phi}R\phi + \frac{\partial L}{\partial(\partial_\mu\phi)}(R\partial_\mu\phi - \partial_\mu\xi^\nu\partial_\nu\phi\right)dV\right) \\ =&\epsilon\int_M\left(L\partial_\mu\xi^\mu + \left(\frac{\partial L}{\partial\phi} - \partial_\mu\left(\frac{\partial L}{\partial(\partial_\mu\phi)}\right)\right)R\phi + \partial_\mu\left(\frac{\partial L}{\partial(\partial_\mu\phi)}R\phi\right) - \frac{\partial L}{\partial(\partial_\mu\phi)}\partial_\mu\xi^\nu\partial_\nu\phi\right)dV. \end{aligned} \tag{11}$$

Therefore, using the Euler-Lagrangian equation, we have

$$\begin{aligned} &\lim_{\epsilon\to 0}\frac{\mathcal{S}(e^{Q\epsilon}M, e^{Q\epsilon}\phi) - \mathcal{S}(M,\phi)}{\epsilon} \\ =&\int_M\left(L\partial_\mu\xi^\mu + \partial_\mu\left(\frac{\partial L}{\partial(\partial_\mu\phi)}R\phi\right) + \left(\frac{\partial L}{\partial\phi} - \partial_\mu\left(\frac{\partial L}{\partial(\partial_\mu\phi)}\right)\right)R\phi - \frac{\partial L}{\partial(\partial_\mu\phi)}\partial_\mu\xi^\nu\partial_\nu\phi\right)dV \\ =&\int_M\left(\partial_\mu\left(L\xi^\mu\right) - \xi^\mu\partial_\mu L + \partial_\mu\left(\frac{\partial L}{\partial(\partial_\mu\phi)}R\phi\right) - \frac{\partial L}{\partial(\partial_\mu\phi)}\partial_\mu\xi^\nu\partial_\nu\phi\right)dV \\ =&\int_M\left(\partial_\mu\left(L\xi^\mu\right) - \xi^\mu\frac{\partial L}{\partial\phi}\partial_\mu\phi - \xi^\mu\frac{\partial L}{\partial(\partial_\nu\phi)}\partial_\mu\partial_\nu\phi + \partial_\mu\left(\frac{\partial L}{\partial(\partial_\mu\phi)}R\phi\right) - \frac{\partial L}{\partial(\partial_\mu\phi)}\partial_\mu\xi^\nu\partial_\nu\phi\right)dV \\ =&\int_M\left(\partial_\mu\left(L\xi^\mu\right) - \xi^\mu\partial_\nu\left(\frac{\partial L}{\partial(\partial_\nu\phi)}\partial_\mu\phi\right) + \partial_\mu\left(\frac{\partial L}{\partial(\partial_\mu\phi)}R\phi\right) - \frac{\partial L}{\partial(\partial_\mu\phi)}\partial_\mu\xi^\nu\partial_\nu\phi\right)dV \\ =&\int_M\partial_\mu\left(L\xi^\mu - \frac{\partial L}{\partial(\partial_\mu\phi)}\xi^\nu\partial_\nu\phi + \frac{\partial L}{\partial(\partial_\mu\phi)}R\phi\right)dV \end{aligned} \tag{12}$$

For the boundary term, using the same trick, up to terms of order $\mathcal{O}(\epsilon^2)$, brutal force computation gives us

$$\begin{aligned}&\int_{e^{Q\epsilon}M}\partial_\mu[F^\mu\left((e^{Q\epsilon}\phi)(p),(\partial_\nu(e^{Q\epsilon}\phi)(p))\right)]dV-\int_M\partial_\mu[F^\mu\left(\phi(p),(\partial_\nu\phi)(p)\right)]dV\\&=\int_M\partial_\mu\left(\frac{\partial F^\mu}{\partial\phi}\left(R\phi-\xi^\nu\partial_\nu\phi\right)+\frac{\partial F^\mu}{\partial(\partial_\nu\phi)}\partial_\nu\left(R\phi-\xi^\nu\partial_\nu\phi\right)+\xi^\mu\partial_\nu F^\nu\right)dV\end{aligned}\tag{13}$$

Using the relation

$$Q\phi=R\phi-\xi^\nu\partial_\nu\phi,\tag{14}$$

then combining Eq. (5), Eq.(12) and Eq.(13), we have

$$\int_M\partial_\mu\left(L\xi^\mu+\frac{\partial L}{\partial(\partial_\mu\phi)}Q\phi+\frac{\partial F^\mu}{\partial\phi}Q\phi+\frac{\partial F^\mu}{\partial(\partial_\nu\phi)}\partial_\nu\left(Q\phi\right)+\xi^\mu\partial_\nu F^\nu\right)dV=0.\tag{15}$$

Since M is arbitrary, we conclude that

$$\partial_\mu\left(L\xi^\mu+\frac{\partial L}{\partial(\partial_\mu\phi)}Q\phi+\frac{\partial F^\mu}{\partial\phi}Q\phi+\frac{\partial F^\mu}{\partial(\partial_\nu\phi)}\partial_\nu\left(Q\phi\right)+\xi^\mu\partial_\nu F^\nu\right)=0.\tag{16}$$

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